RES POLYTECHNIC, BANGALORE-91

CIE Theory Test III

Program	Mechanical Engineering	**Semester - III**
Course Name	Programming Concepts in Mechanical Engineering	**Test III**
Course Code	25ME34I	**Marks - 50**
Name of the Course Coordinator	Vinay Kumar	**Duration - 90 min**

Note: Answer any one full question from each section. Each question carries equal marks.

Q. No	Questions	CL	CO	Marks
	SECTION I			
1	a) Write a Python program that stores the names of five students in a list, adds one new student, removes one student and then displays the list in sorted order. Use the append(), remove() and sort() methods.	L3	CO1	7
	b) Write a Python program to read a text file named data.txt and count the number of words in it. Use try, except and finally blocks to handle the situation when the file is not found.	L3	CO2	6
	c) Write a Python class Motor with the attributes model, rated power (kW) and colour. Use the __init__ method and include methods to start, stop and display the motor details. Create one object and call all the methods.	L3	CO2	6
	d) Explain how a DC motor is controlled using an L298N motor driver and Raspberry Pi. List the GPIO connections required and write a short Python program to run the motor for 5 seconds and then stop it.	L3	CO4	6
	OR			
2	a) Create a Python dictionary to store the record (name, age and grade) of a student. Write functions to add a new student record and to remove a record, and use a loop to display all the records.	L3	CO1	7
	b) Write a Python program that writes ten random sensor readings to a file and then reads them back. Explain the purpose of the modes "w", "r" and "a" used in the open() function.	L3	CO2	6
	c) Write a Python program with a base class Machine and derived classes Lathe and Drill. Show inheritance and method overriding by giving each derived class its own operate() method, and explain how overriding works.	L3	CO2	6
	d) Explain how a DHT11 temperature-humidity sensor and a PIR motion sensor are interfaced with Raspberry Pi. Write a short Python program that switches ON a cooling fan connected to a GPIO pin when the temperature is above 35 °C.	L3	CO4	6

Q. No	Questions	CL	CO	Marks
	SECTION II			
3	a) Five samples, each containing four shafts, are measured for diameter (in mm). Write a Python program to calculate the sample average (X-bar) and the sample range (R) of each sample, and plot the X-bar chart and the R chart using Matplotlib.	L3	CO3	7
	b) Write Python code to calculate the mean, variance and standard deviation of ten measured shaft diameters. Using USL = 25.5 mm and LSL = 24.5 mm, calculate the process capability Cp and the process capability index Cpk.	L3	CO2	6
	c) Explain hypothesis testing in quality control. Distinguish between the t-test and ANOVA with an example from a workshop, such as comparing the hardness of components from two or three batches.	L3	CO3	6
	d) Explain the general steps of the finite element method. State any three engineering applications and any two advantages of the method.	L3	CO3	6
	OR			
4	a) The numbers of defective pieces found in 10 samples of 50 pieces each are 4, 3, 5, 2, 6, 4, 3, 7, 2 and 4. Write a Python program to calculate the fraction defective, the centre line, UCL and LCL, and plot the p-chart using Matplotlib.	L3	CO3	7
	b) The sample averages of 8 samples are 25.0, 25.1, 24.9, 25.5, 25.0, 24.5, 25.2 and 25.1 mm, with UCL = 25.4 mm and LCL = 24.6 mm. Write Python code to identify and print the in-control and out-of-control samples.	L3	CO2	6
	c) The defects found in a casting shop are: blowholes 40, cracks 25, misruns 15, shrinkage 10 and others 10. Write Python code to draw a Pareto chart and explain how it helps in continuous improvement.	L3	CO3	6
	d) Write the stiffness matrix of a bar element of area A, length L and modulus of elasticity E. For a bar made of two elements, show how the global stiffness matrix is assembled and state any two properties of the stiffness matrix.	L3	CO3	6

Signature of the Course Coordinator

Signature of the HOD

Signature of the IQAC Chairman